



Department of Computer Science and Engineering (Data Science)

COURSE CODE: DJ19SDL303

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COURSE NAME: FDA

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DIVISION:B

SAP ID: 60009210191

Exp- 4

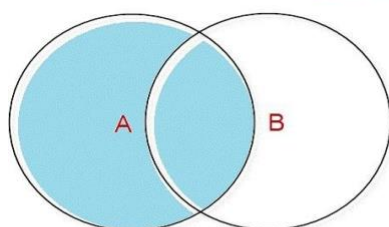
(Data Visualization)

Aim: Perform joins, blends and create dual axis charts and table calculations, bins, distributions and create Heat maps

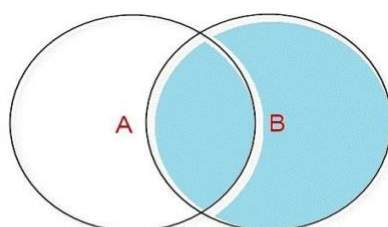
Theory:

Data joining is a common requirement in any data analysis. You may need to join data from different tables in a single source or join data from multiple sources. A join means combining columns from one or more tables in a relational database. It also creates a set that can be saved as a table, or it can be used as it is. Joins are a concept taken from relational databases like SQL and may be very useful in the future.

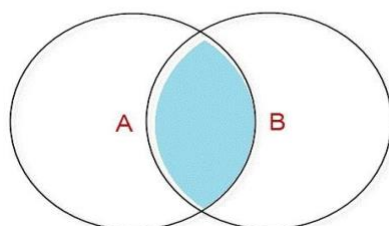
Common SQL Joins



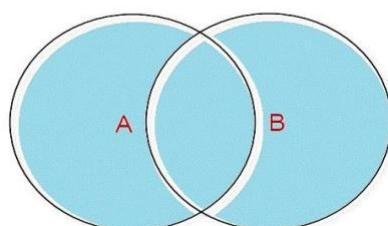
Left Outer



Right Outer



Inner



Full Outer

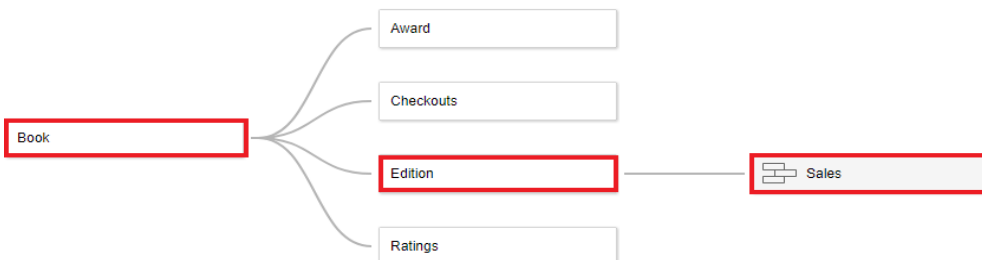
Types of joins.

A blend is a smart join on the go. It is always a left join. Blends work for databases of different sources as well. Tableau may automatically suggest/use blends if fields are having the same name. If not, fields can be blended manually. Tableau Relationships



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Relationships are a flexible way to combine data for multi-table analysis in Tableau. Think of a relationship as a contract between two tables. When you are building a viz with fields from these tables, Tableau brings in data from these tables using that contract to build a query with the appropriate joins.



When to use Joins & Blends:

Joins: Combining data at row level

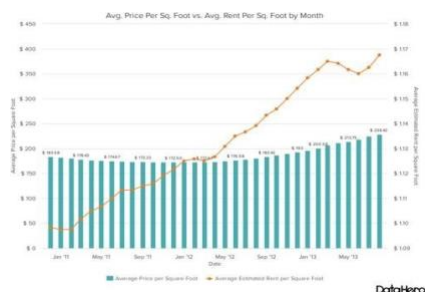
Blends: When data sources have different level of granularity or when data sources are from different sources

Relationships: Tableau connects the data at the right level of aggregation. It is a sort of smart on the fly method of connecting tables. It's more flexible than a join. Whenever in doubt, use a relationship. Use a Join or blend if 100% sure that it is required.

Dual Axis Charts:

A dual axis chart is a great way to easily illustrate the relationship between two different variables. They illustrate a lot of information with limited space and allow you to discover trends you may have otherwise missed if you're switching between graphs. Synchronizing Dual Axis charts is very important in most cases.

Examples of Dual Axis Charts





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Table Calculations:

A table calculation is a transformation you apply to the values in a visualization. Table calculations are a special type of calculated field that computes on the local data in Tableau. They are calculated based on what is currently in the visualization and do not consider any measures or dimensions that are filtered out of the visualization.

You can use table calculations for a variety of purposes, including:

- Transforming values to rankings
- Transforming values to show running totals

Bins:

Any discrete field in Tableau can be considered as a set of bins. For example, suppose you create a view with **Profit** on **Rows** and **State** on **Columns**. You could consider the **State** field as a set of bins—each profit value is sorted into a bin corresponding to the state from which the value was recorded. But if you want to see values for **Profit** assigned to bins without reference to a dimension, you can create a numeric bin, with each individual bin corresponding to a range of values.

Heat Maps:

A heat map is a two-dimensional representation of data in which values are represented by colors. A simple heat map provides an immediate visual summary of information. More elaborate heat maps allow the viewer to understand complex data sets.

There can be many ways to display heat maps, but they all share one thing in common -- they use color to communicate relationships between data values that would be much harder to understand if presented numerically in a spreadsheet.

In the United States, many people are familiar with heat maps from viewing television news programs. During a presidential election, for instance, a geographic heat map with the colors red and blue will quickly inform the viewer which states each candidate has won.





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Lab Assignments to complete in this session

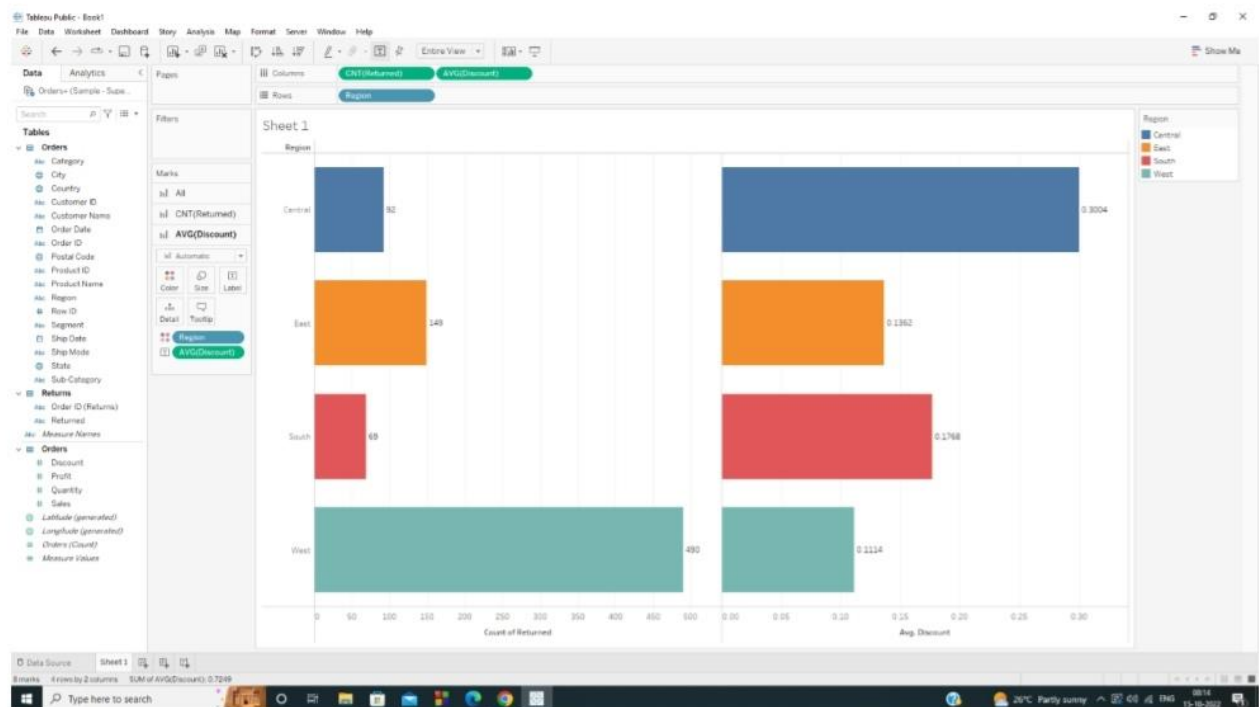
1. Use the given datasets and perform the following tasks:

Datasets:

<https://community.tableau.com/s/question/0D54T00000CWeX8SAL/sample-superstore-sales-excelxls>

<https://data.world/2918diy/coffee-chain/workspace/file?filename=Coffee+Chain.txt>

- a) The company wants to analyze region wise return of products from superstore dataset. Is there any relationship of return of product with the discounts offered on it?

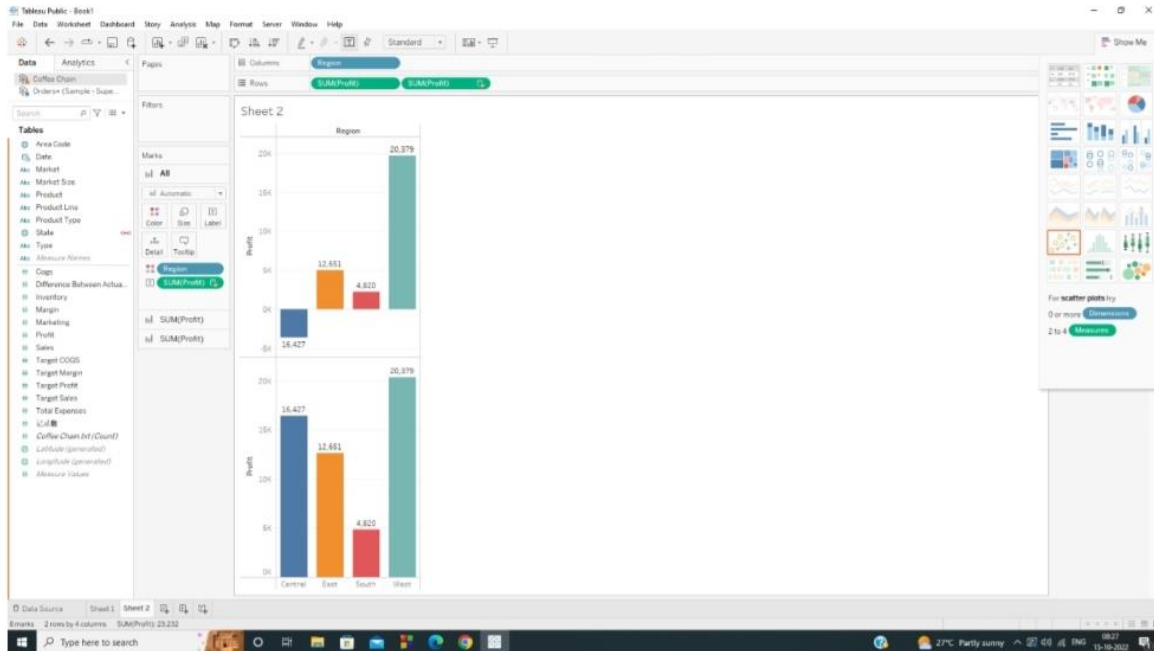


Analysis: There is no visible relationship between discount and returned item. However, we could note from the above graph that number of items returned increases drastically when discount is more than 60% (i.e. 54.60%)

- b) Compare the region wise profits of superstore and coffee chain (Market means region). Find which region has performed well for both.



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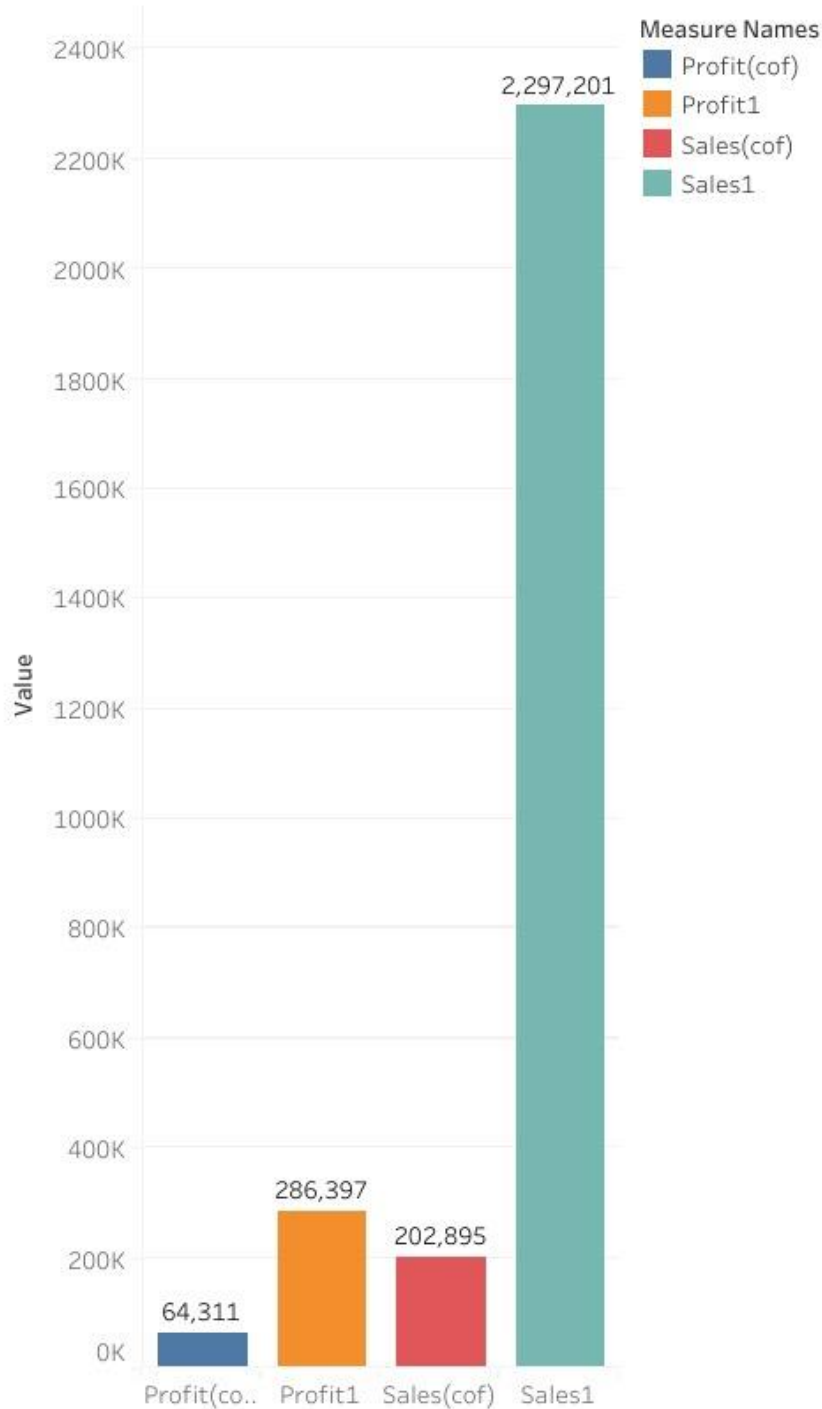
Analysis: Western region has performed well for both.

c) Show sales and profit together for both the datasets choosing an appropriate visualization.



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Sheet 1



d) Analyze year wise profit for both the datasets and illustrate the limitations of blending.



Shri Vile Parle Kelavani Mandal's

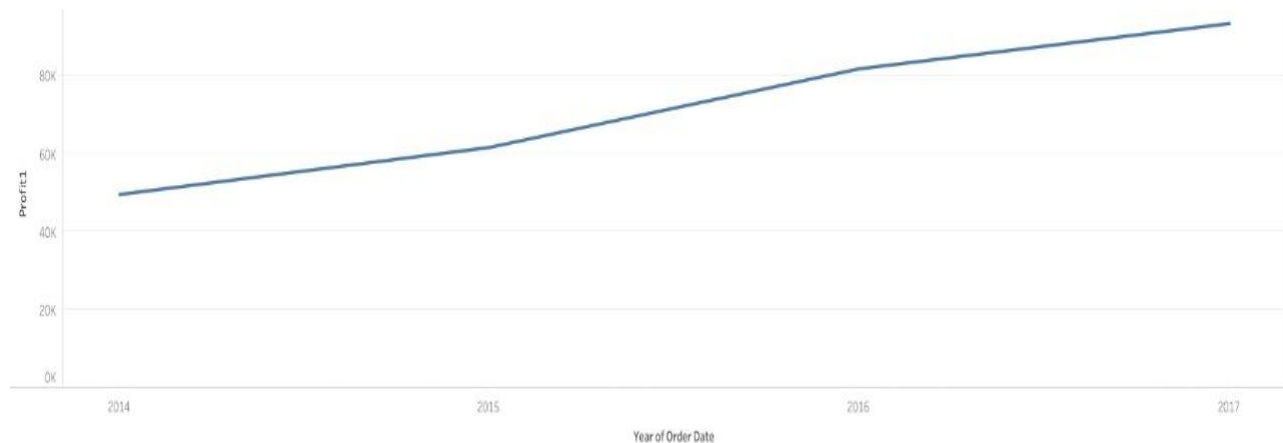
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(Autonomous College Affiliated to the University of Mumbai)

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ANALYSIS: We could analyse the profit in time interval(i.e YEAR-2012 to 2013) effectively by using a line graph. We could comment from the above graph that profit generated by Cofee Mart remains same from 2012 to 2013 but profit generated by Coffee Chain increases from 2012 to 2013

Limitations of blending are-

- Cube data sources (datacubes) can only be used as the primary data source for data blending
- Cannot publish a data blending source as a single server data source
- Tableau times out when working with blended data
- A Level of Detail expression is being taken from the secondary data source
- A published data source is the primary data source